Building and evaluation of a PBPK model for (E)-Clomiphene in CYP2D6 activity-score groups

Version	vevaluation
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/releases/tag/v1.0
OSP Version	evaluation
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This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

(E)-Clomiphene is the trans-isomer of clomiphene and contributes to the clinical pharmacology of clomiphene citrate, a selective estrogen receptor modulator used in reproductive medicine. Its pharmacokinetics are complex because clomiphene is administered as an E/Z mixture, undergoes sequential oxidative metabolism, and shows clinically relevant variability linked to CYP2D6 activity Kovar 2022.

The model describes (E)-clomiphene as a whole-body parent-metabolite PBPK model with (E)-N-desethylclomiphene, (E)-4-hydroxyclomiphene and (E)-4-hydroxy-N-desethylclomiphene as measured metabolites. This structure supports evaluation of parent and metabolite plasma concentration-time data after oral clomiphene or enclomiphene administration and enables activity-score-dependent CYP2D6 simulations.

The clinical evaluation covers adult plasma pharmacokinetic data after single-dose and multiple-dose oral administration. The dataset includes a CYP2D6 activity-score-stratified panel study with parent and metabolite measurements and external literature studies reporting (E)-clomiphene concentration-time data. The model was published by Kovar 2022 and was subsequently integrated into the CYP2D6 interaction network by Rüdesheim 2025.

2 Methods

2.1 Modeling strategy

The PBPK model was developed using the Open Systems Pharmacology Suite. Distribution and elimination processes were implemented in a whole-body model structure using standard physiological organ compartments and compound-specific physicochemical and biochemical input parameters.

Model development followed an iterative workflow in which in vitro and physicochemical information was combined with clinical plasma concentration-time data. Parameters with direct literature support were used as fixed inputs where possible, while selected parameters controlling oral absorption, metabolism, residual clearance and activity-score-dependent CYP2D6 scaling were optimized against clinical data.

The model represents (E)-clomiphene and three metabolites as separate compounds. (E)-N-Desethylclomiphene is included as the main desethyl metabolite, while (E)-4-hydroxyclomiphene and (E)-4-hydroxy-N-desethylclomiphene are included as active hydroxylated metabolites. The four-compound structure is required because the available panel study reports parent and metabolite concentration-time data across CYP2D6 activity-score groups.

The clinical panel study by Mürdter 2016 informed activity-score-dependent parent-metabolite disposition after oral clomiphene citrate administration. Literature studies by Mikkelson 1986, Ratiopharm GmbH 2016, Wiehle 2013, and Miller 2019 provided additional single-dose and multiple-dose oral concentration-time data for (E)-clomiphene. The literature studies evaluate oral formulation behavior and the parent compound over a broader dose range.

The major proteins represented in the model are CYP2D6, CYP3A4 and CYP2B6. CYP2D6 accounts for activity-score-dependent formation and elimination pathways. CYP3A4 contributes to desethylation and metabolite turnover, while CYP2B6 contributes to (E)-4-hydroxyclomiphene formation. Renal filtration, enterohepatic recirculation and residual hepatic clearance are retained where required to describe parent and metabolite disposition.

2.2 Data used

In vitro and physicochemical data

Drug-dependent parameters for (E)-clomiphene and its metabolites were taken from the published PBPK model and its supplement (Kovar 2022). Table 1 gives the physicochemical, distribution, absorption, metabolism, and elimination parameters used in the final model. The Source column identifies the original literature or database named in Kovar 2022. It identifies model-fitted and assumed values directly.

Parameter	Unit	Value	Source	Description
(E)-Clomiphene				
MW	g/mol	405.96	Siramshetty 2022	Molecular weight.
pK_a,base	-	9.31	ChemAxon 2009	pKa of a basic ionization site.
Solubility	mg/mL	0.0138	Das 2020	Aqueous solubility at pH 6.8.
logP	-	5.67	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	0.08	Optimized	Fraction unbound in plasma.
$K_{m,CYP2D6}$	µmol/L	0.13	Kröner 2018	Michaelis constant for formation of (E)-4-hydroxyclomiphene from (E)-clomiphene.
$K_{m,CYP2D6}$	µmol/L	0.03	Kröner 2018	Michaelis constant for formation of other metabolite from (E)-clomiphene.
$K_{m,CYP2D6}$	µmol/L	0.78	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of (E)-N-desethylclomiphene from (E)-clomiphene.
$K_{m,CYP3A4}$	µmol/L	0.78	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of (E)-N-desethylclomiphene from (E)-clomiphene.
$k_{cat,CYP3A4}$	1/min	45.0	Optimized	Catalytic rate constant for formation of (E)-N-desethylclomiphene from (E)-clomiphene.
$K_{m,CYP2B6}$	µmol/L	0.60	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of (E)-4-hydroxyclomiphene from (E)-clomiphene.
$k_{cat,CYP2B6}$	1/min	7.5	Optimized	Catalytic rate constant for formation of (E)-4-hydroxyclomiphene from (E)-clomiphene.
GFR fraction	-	0.92	Optimized	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed	Fraction released continuously into bile.
Partition coefficients	-	Schmitt	Schmitt 2008	Tissue-to-plasma partition coefficients calculated with the Schmitt method.
Cellular permeabilities	-	Charge-dependent Schmitt	Kawai 1994	Cellular permeabilities calculated with the charge-dependent Schmitt method.
P_{int}	cm/min	0.08	Optimized	Specific transcellular intestinal permeability.

Parameter	Unit	Value	Source	Description
t_{50}	min	6.80	Assumed	Time to dissolve 50% of the dose.
Weibull shape	-	0.47	Assumed	Shape parameter of the Weibull dissolution function.
(E)-N-desethylclomiphene				
MW	g/mol	377.91	ChemAxon 2009	Molecular weight.
pK_a,base	-	8.14	Optimized	pKa of a basic ionization site.
Solubility	mg/mL	0.46	ChemAxon 2009	Aqueous solubility at pH 6.5.
logP	-	4.17	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	0.86	Optimized	Fraction unbound in plasma.
$K_{m,CYP2D6}$	µmol/L	0.49	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of (E)-4-hydroxy-N-desethylclomiphene from (E)-N-desethylclomiphene.
$K_{m,CYP2D6}$	µmol/L	0.97	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of other metabolite from (E)-N-desethylclomiphene.
$K_{m,CYP3A4}$	µmol/L	0.97	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of other metabolite from (E)-N-desethylclomiphene.
$k_{cat,CYP3A4}$	1/min	0.8	Optimized	Catalytic rate constant for formation of other metabolite from (E)-N-desethylclomiphene.
GFR fraction	-	0.10	Optimized	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed	Fraction released continuously into bile.
Partition coefficients	-	Rodgers and Rowland	Rodgers 2005; Rodgers 2006	Tissue-to-plasma partition coefficients calculated with the Rodgers and Rowland method.
Cellular permeabilities	-	Charge-dependent Schmitt	Kawai 1994	Cellular permeabilities calculated with the charge-dependent Schmitt method.
(E)-4-hydroxyclomiphene				
MW	g/mol	421.97	ChemAxon 2009	Molecular weight.

Parameter	Unit	Value	Source	Description
pK_a,acid	-	8.64	ChemAxon 2009	pKa of an acidic ionization site.
pK_a,base	-	7.90	Optimized	pKa of a basic ionization site.
Solubility	mg/mL	0.06	ChemAxon 2009	Aqueous solubility at pH 6.5.
logP	-	5.50	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	0.45	Optimized	Fraction unbound in plasma.
$K_{m,CYP2D6}$	µmol/L	3.60	Kröner 2018	Michaelis constant for formation of other metabolite from (E)-4-hydroxyclomiphene.
$K_{m,CYP3A4}$	µmol/L	3.40	Mürdter 2012; Ganchev 2014	Michaelis constant for formation of (E)-4-hydroxy-N-desethylclomiphene from (E)-4-hydroxyclomiphene.
$k_{cat,CYP3A4}$	1/min	19.5	Optimized	Catalytic rate constant for formation of (E)-4-hydroxy-N-desethylclomiphene from (E)-4-hydroxyclomiphene.
CL_{hep}	1/min	23.78	Optimized	Unspecific hepatic clearance.
GFR fraction	-	0.24	Optimized	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed	Fraction released continuously into bile.
Partition coefficients	-	Berezhkovskiy	Berezhkovskiy 2004	Tissue-to-plasma partition coefficients calculated with the Berezhkovskiy method.
Cellular permeabilities	cm/min	2.23	Open Systems Pharmacology Suite Community 2021	Calculated cellular permeability.
(E)-4-hydroxy-N-desethylclomiphene				
MW	g/mol	393.91	ChemAxon 2009	Molecular weight.
pK_a,acid	-	8.69	ChemAxon 2009	pKa of an acidic ionization site.
pK_a,base	-	9.65	ChemAxon 2009	pKa of a basic ionization site.
Solubility	mg/mL	0.17	ChemAxon 2009	Aqueous solubility at pH 6.5.

Parameter	Unit	Value	Source	Description
logP	-	3.71	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	1.32	Watanabe 2018	Fraction unbound in plasma.
$K_{m,CYP2D6}$	µmol/L	8.86	Assumed	Michaelis constant for formation of other metabolite from (E)-4-hydroxy-N-desethylclomiphene.
CL_{hep}	1/min	8.50	Optimized	Unspecific hepatic clearance.
GFR fraction	-	0.13	Optimized	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed	Fraction released continuously into bile.
Partition coefficients	-	Schmitt	Schmitt 2008	Tissue-to-plasma partition coefficients calculated with the Schmitt method.
Cellular permeabilities	-	Charge-dependent Schmitt	Kawai 1994	Cellular permeabilities calculated with the charge-dependent Schmitt method.

Table 1: Drug-dependent parameters used in the final clomiphene model. Original sources are those assigned in Kovar 2022. Other metabolite: an undefined downstream product represented as an elimination pathway in the model.

The CYP2D6 activity-score-specific catalytic rates are listed in Table 1a.

Substrate	Product or pathway	AS = 0	AS = 0.5	AS = 0.75	AS = 1	AS = 2	AS = 3	Unit
(E)-Clomiphene	(E)-4-hydroxyclomiphene	0	57.48	81.57	175.08	306.38	467.23	1/min
(E)-Clomiphene	(E)-N-desethylclomiphene	0	21.12	27.78	61.56	121.36	171.54	1/min
(E)-Clomiphene	Other CYP2D6 hydroxylation	0	22.68	29.84	66.13	130.35	184.25	1/min
(E)-N-desethylclomiphene	(E)-4-hydroxy-N-desethylclomiphene	0	10.35	12.37	28.59	64.52	84.01	1/min
(E)-N-desethylclomiphene	Other CYP2D6 desethylation	0	1.02	1.34	2.96	5.84	8.25	1/min
(E)-4-hydroxyclomiphene	Other CYP2D6 elimination	0	148.80	195.78	433.82	855.17	1208.79	1/min
(E)-4-hydroxy-N-desethylclomiphene	Other CYP2D6 elimination	0	36.84	48.47	107.40	211.71	299.26	1/min

Table 1a: CYP2D6 activity-score-specific k_{cat} values. AS = 2 values were optimized. AS = 0 values were assumed to be zero. Other values were calculated from AS = 2 with the pathway-specific *in vitro* scaling factors in Supplement Table S8 of Kovar 2022. AS: activity score.

External studies did not report CYP2D6 activity scores or phenotypes. The model used the study-specific optimized CYP2D6 catalytic rates in Table 1b.

Study profile	(E)-Clomiphene to (E)-4-hydroxyclomiphene	(E)-Clomiphene to other metabolite	(E)-Clomiphene to (E)-N-desethylclomiphene	Unit
Mikkelson 1986	213.0	90.6	84.4	1/min
Miller 2019	18.1	7.7	7.2	1/min
Ratiopharm GmbH 2016	283.1	120.5	112.1	1/min
Wiehle 2013, 6.25 mg	87.7	37.3	34.8	1/min
Wiehle 2013, 12.5 mg	124.1	52.8	49.1	1/min
Wiehle 2013, 25 mg	43.3	18.4	17.1	1/min

Table 1b: Study-specific optimized CYP2D6 k_{cat} values for the external clinical profiles, as reported in Kovar 2022.

Clinical data

The evaluation includes 12 plasma concentration-time profiles after oral clomiphene or enclomiphene administration (Table 2). The AS = 0 and AS = 2 groups of the Mürdter 2016 panel were used for model building. The AS = 0.5, 0.75, 1 and 3 groups of the panel and six external profiles were used for model verification. For each external profile, the CYP2D6 k_{cat} values were fitted to that profile (Table 1b).

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Mikkelson 1986	50 clomiphene citrate (21 (E)-clomiphene model dose), oral	32	62.4	Female	23	Tablet	Study-specific CYP2D6 k_{cat}
Miller 2019	50 clomiphene citrate (21 (E)-clomiphene model dose), oral, once daily for 30 days (model schedule)	31.5 ± 3.6	77.9 ± 8.2	Male	12	Tablet	Study-specific CYP2D6 k_{cat}
Mürdter 2016+	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	25.2 (22–29)	62.3 (50–70)	Female	6	Tablet	AS = 0 (PM)
Mürdter 2016	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	24.3 (21–30)	59.3 (55.5–64)	Female	4	Tablet	AS = 0.5 (IM)
Mürdter 2016	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	22	63	Female	1	Tablet	AS = 0.75 (IM)
Mürdter 2016	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	25.5 (23–28)	68.8 (63.5–74)	Female	2	Tablet	AS = 1 (IM)
Mürdter 2016+	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	32.3 (26–43)	56.5 (48–63.5)	Female	3	Tablet	AS = 2 (NM)
Mürdter 2016	100 clomiphene citrate (42 (E)-clomiphene model dose), oral	25.7 (22–28)	61.7 (54–73)	Female	3	Tablet	AS = 3 (UM)
Ratiopharm GmbH 2016	50 clomiphene citrate (21 (E)-clomiphene model dose), oral	NR	NR	NR	18	Tablet	Study-specific CYP2D6 k_{cat}
Wiehle 2013	6.25 enclomiphene citrate (4.24 (E)-clomiphene model dose), oral, once daily for 43 days (model schedule)	53.3 ± 10.2	NR	Male	16	Capsule	Study-specific CYP2D6 k_{cat}
Wiehle 2013	12.5 enclomiphene citrate (8.48 (E)-clomiphene model dose), oral, once daily for 43 days (model schedule)	53.3 ± 10.2	NR	Male	14	Capsule	Study-specific CYP2D6 k_{cat}
Wiehle 2013	25 enclomiphene citrate (16.97 (E)-clomiphene model dose), oral, once daily for 43 days (model schedule)	53.3 ± 10.2	NR	Male	16	Capsule	Study-specific CYP2D6 k_{cat}

Table 2: Clinical (E)-clomiphene concentration-time profiles used for model building and verification. Data for the external studies are from Kovar 2022. The model-dose equivalents identify the evaluated simulation dose. *: Single oral dose unless otherwise specified. Age and weight are means. Values with ± are mean ± SD. Values in parentheses are ranges. AS: activity score. IM: intermediate metabolizer. NM: normal metabolizer. NR: not reported. For Ratiopharm GmbH 2016, the simulation used the default 30-year-old European male individual because the study did not report demographics. PM: poor metabolizer. PT: predicted phenotype. UM: ultrarapid metabolizer. $^{+}$: data used for model building. Parenthetical PTs for AS-coded rows use the CYP2D6 activity score-to-phenotype mapping from Moore 2026. Study-specific CYP2D6 k_{cat}: activity score and phenotype were not reported, and the study-specific optimized values in Table 1b were used.

2.3 Model parameters and assumptions

Absorption

The model includes oral administration of (E)-clomiphene after clomiphene citrate or enclomiphene citrate dosing. The clinical panel study used clomiphene citrate tablets containing the E/Z isomer mixture, and the model evaluates the (E)-clomiphene fraction relevant to the measured parent-metabolite dataset (Kovar 2022).

Oral absorption is represented with an assumed Weibull dissolution function and compound intestinal permeability. Dissolution profiles were not available. The same Weibull function is used for the clomiphene citrate tablets and for the enclomiphene citrate capsules of Wiehle 2013 (Kovar 2022). The formulation parameters describe the oral input profile, while differences between CYP2D6 activity-score groups are assigned to metabolism rather than absorption.

The external literature studies include single-dose and multiple-dose oral dosing conditions. These data support evaluation of oral input behavior across lower single doses and repeated enclomiphene exposure, but they do not by themselves resolve individual CYP2D6 activity in the absence of genotype or phenotype information.

Distribution

(E)-Clomiphene is represented as a highly lipophilic, highly protein-bound compound with logP of 5.67 and f_u of 0.08% as summarized in Section 2.2. The metabolites are less lipophilic but remain substantially bound, with logP values of 4.17 for (E)-N-desethylclomiphene, 5.50 for (E)-4-hydroxyclomiphene, and 3.71 for (E)-4-hydroxy-N-desethylclomiphene (Kovar 2022).

Partition coefficients were calculated with the Schmitt, Rodgers and Rowland, or Berezhkovskiy methods, depending on the compound (Schmitt 2008, Rodgers 2005, Rodgers 2006, Berezhkovskiy 2004). Cellular permeabilities were calculated with the Charge-dependent Schmitt or PK-Sim Standard methods. The model therefore treats the parent and metabolites as separate distribution entities rather than as a single lumped active moiety.

The active hydroxylated metabolites are represented with their own molecular weight, pK_a, lipophilicity, plasma binding and clearance parameters. This separation is required because metabolite exposure is controlled by both formation from upstream compounds and metabolite-specific elimination.

Metabolism and elimination

Clomiphene elimination is represented by CYP2D6-dependent hydroxylation and desethylation, CYP3A4-dependent desethylation and metabolite turnover, CYP2B6-dependent hydroxylation, renal filtration, enterohepatic recirculation and residual hepatic clearance components.

- CYP2D6

CYP2D6 forms (E)-4-hydroxyclomiphene from (E)-clomiphene and contributes to desethylation and downstream metabolite elimination. Pathway-specific K_m values are used, while k_{cat} values depend on activity score (Kovar 2022, Mürdter 2012).

The CYP2D6 activity-score implementation is the key determinant of simulated exposure differences across poor, intermediate, normal and ultrarapid metabolizer groups. Poor-metabolizer activity is set to zero, while non-zero activity-score groups use scaled k_{cat} values.

- CYP3A4, CYP2B6 and residual clearance

CYP3A4 is implemented for (E)-N-desethylclomiphene formation and downstream metabolite turnover. CYP2B6 contributes to (E)-4-hydroxyclomiphene formation from (E)-clomiphene. These pathways support non-CYP2D6 clearance and metabolite formation where the *in vitro* data indicate additional enzymatic contribution (Kovar 2022, Mazzarino 2013).

Residual hepatic clearance terms are empirical. They should be interpreted as structural model components required to describe total disposition rather than as direct measurements of a single biochemical pathway.

- Renal filtration and enterohepatic recirculation

Renal filtration is included with compound-specific GFR fractions for the parent and metabolites. Enterohepatic recirculation is represented with continuous bile release fractions where required by the model structure.

3 Results and Discussion

The PBPK model for (E)-clomiphene was developed and evaluated with clinical pharmacokinetic data after oral clomiphene or enclomiphene administration. The evaluation covers single-dose and multiple-dose administration, (E)-clomiphene and three circulating metabolites, and CYP2D6 activity scores from 0 to 3.

Model-building data from the AS = 0 and AS = 2 groups of Mürdter 2016 supported the parent-metabolite structure and the CYP2D6 k_{cat} values at AS = 2. Verification used the AS = 0.5, 0.75, 1 and 3 groups of Mürdter 2016, with k_{cat} values extrapolated from AS = 2 with in vitro scaling factors (Kovar 2022), and (E)-clomiphene profiles from Mikkelson 1986, Miller 2019, Ratiopharm GmbH 2016, and Wiehle 2013. The CYP2D6 k_{cat} values of each external profile were fitted to that profile. These profiles are therefore not independent of the model parameters. The external studies mainly assess parent-compound disposition, while the activity-score panel assesses parent-metabolite behavior across CYP2D6 groups.

The model includes CYP2D6-dependent hydroxylation and desethylation, CYP3A4- and CYP2B6-mediated metabolism, passive renal filtration, residual hepatic clearance, and enterohepatic recirculation. Goodness-of-fit diagnostics are separated for (E)-clomiphene, (E)-N-desethylclomiphene, (E)-4-hydroxyclomiphene, and (E)-4-hydroxy-N-desethylclomiphene to avoid ambiguous parent-metabolite legends.

The next sections show:

1. the final model input parameters for the building blocks in Section 3.1.
2. analyte-specific goodness-of-fit diagnostics in Section 3.2.
3. simulated vs. observed concentration-time profiles for model building and verification in Section 3.3.

Residual uncertainty concerns non-CYP2D6 clearance, protein-binding estimates, and individual variability within the same CYP2D6 activity-score group. Further genotype-specific refinement would require additional clinical data with paired parent-metabolite profiles.

3.1 Clomiphene final input parameters

The following input-parameter tables summarize the final model parameterization for (E)-clomiphene and its metabolites. Parameter interpretation should be based on the source descriptions in Section 2.2 and the assumptions in Section 2.3.

Compound: (E)-clomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	13.8 mg/l	Publication-Das et al. 2020	Das et al. 2020 pH 6.8	True
Reference pH	6.8	Publication-Das et al. 2020	Das et al. 2020 pH 6.8	True
Lipophilicity	5.67 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.00084694454938	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Specific intestinal permeability (transcellular)	0.0828652784 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
CI	1	Publication-Siramshetty et al. 2022		
Is small molecule	Yes			
Molecular weight	405.96 g/mol	Publication-Siramshetty et al. 2022		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Schmitt
Cellular permeabilities	Charge dependent Schmitt

Processes

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012

Molecule: CYP3A4

Metabolite: (E)-Deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	44.9466036298 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	306.3819989852 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.9183035142	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2B6-Mürdter et al. 2012

Molecule: CYP2B6

Metabolite: 4-Hydroxy-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	39 pmol/mg mic. protein	Unknown
Km	0.600817495 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	7.493654322 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=2) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	130.3516139745 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=3) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	171.5386005 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.5) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	22.68117655 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.75) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	29.84299905 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=1) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	66.12599871 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=3) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	184.2542995 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=2) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	121.3558299218 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0.5) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	21.11591042 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0.75) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	27.7834835 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=1) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	61.56253235 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	57.48197455 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	81.56688504 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	175.0793273 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	467.2288197 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Miller 2019) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	7.1740592157 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Miller 2019) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	7.7058530943 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Miller 2019) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	18.1120478905 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 12.5 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	52.779552089 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 12.5 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	49.1371464562 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 12.5 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	124.0545029059 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 25 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	18.4144257821 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 6.25 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	37.3233521148 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Study Ratiopharm GmbH 2016) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	120.4548809775 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Mikkelson 1986) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	90.6375098012 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 25 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	17.143615297 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 6.25 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	34.7476048301 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Study Ratiopharm GmbH 2016) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	112.1420871092 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Mikkelson 1986) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	84.3824628525 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 25 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	43.2817700468 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 6.25 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	87.725827714 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Study Ratiopharm GmbH 2016) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	283.1204470445 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Mikkelson 1986) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	213.0368822389 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.46 mg/ml	Database-Chemicalize	Measurement	True
Reference pH	6.5	Database-Chemicalize	Measurement	True
Lipophilicity	4.1726909096 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.0085575590017	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Cl	1			
Is small molecule	Yes			
Molecular weight	377.91 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	Charge dependent Schmitt

Processes

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	84.00858577 1/min	Other-Other-Calculated

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.1005414757	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	8.248111562 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012 - (E)-N,N-didesethylclomiphene

Molecule: CYP3A4

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0.7771909271 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	1.01531891 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	1.335916645 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	2.960118793 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	5.8351672527 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	64.5203767233 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	10.34807199 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	12.36587928 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	28.59140514 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Other-Assumption-Assumed

Compound: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.06 mg/ml		Measurement	True
Reference pH	6.5		Measurement	True
Lipophilicity	5.5012508036 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.00449	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Cl	1			
Is small molecule	Yes			
Molecular weight	421.97 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012

Molecule: CYP3A4

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	161.2 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	3.400160633 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	19.52 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.2386673303	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=3)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	1208.794021 1/min	Other-Other-Calculated

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.0133	
Lipophilicity (experiment)	4.6502520184 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	23.7771867087 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.5)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	148.7990819 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.75)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	195.7839731 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=1)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	433.8173497 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=2)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	855.1672990338 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: (E)-4-hydroxy-N-desethylclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.17 mg/ml	Database-Chemicalize	Measurement	True
Reference pH	6.5	Database-Chemicalize	Measurement	True
Lipophilicity	3.7147097639 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.0132	Publication-Other-Watanabe et al. 2018	Measurement	True
CI	1			
Is small molecule	Yes			
Molecular weight	393.91 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Schmitt
Cellular permeabilities	Charge dependent Schmitt

Processes

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.0132	
Lipophilicity (experiment)	6 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	8.5014115561 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.1294297124	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=3)

Molecule: CYP2D6

Metabolite: (Z)-3,4-Dihydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	299.2603852 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Assumption (AS=0)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-Assumption (AS=0.5)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	36.83809632 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Assumption (AS=0.75)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	48.47011666 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Assumption (AS=1)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	107.3998919 1/min	Other-Other-Calculated

Metabolizing Enzyme: CYP2D6-Assumption (AS=2)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	211.7132372293 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics are separated by analyte. This keeps the parent and three metabolite legends and error measures distinct.

3.2.1 (E)-Clomiphene goodness-of-fit diagnostics

(E)-Clomiphene GOF diagnostics include the external literature studies and the CYP2D6 activity-score panel study. These diagnostics assess parent-compound exposure across oral single-dose, oral multiple-dose, and CYP2D6-stratified conditions.

Table 3-1: GMFE for (E)-Clomiphene goodness-of-fit diagnostics

Group	GMFE
(E)-clomiphene	1.45

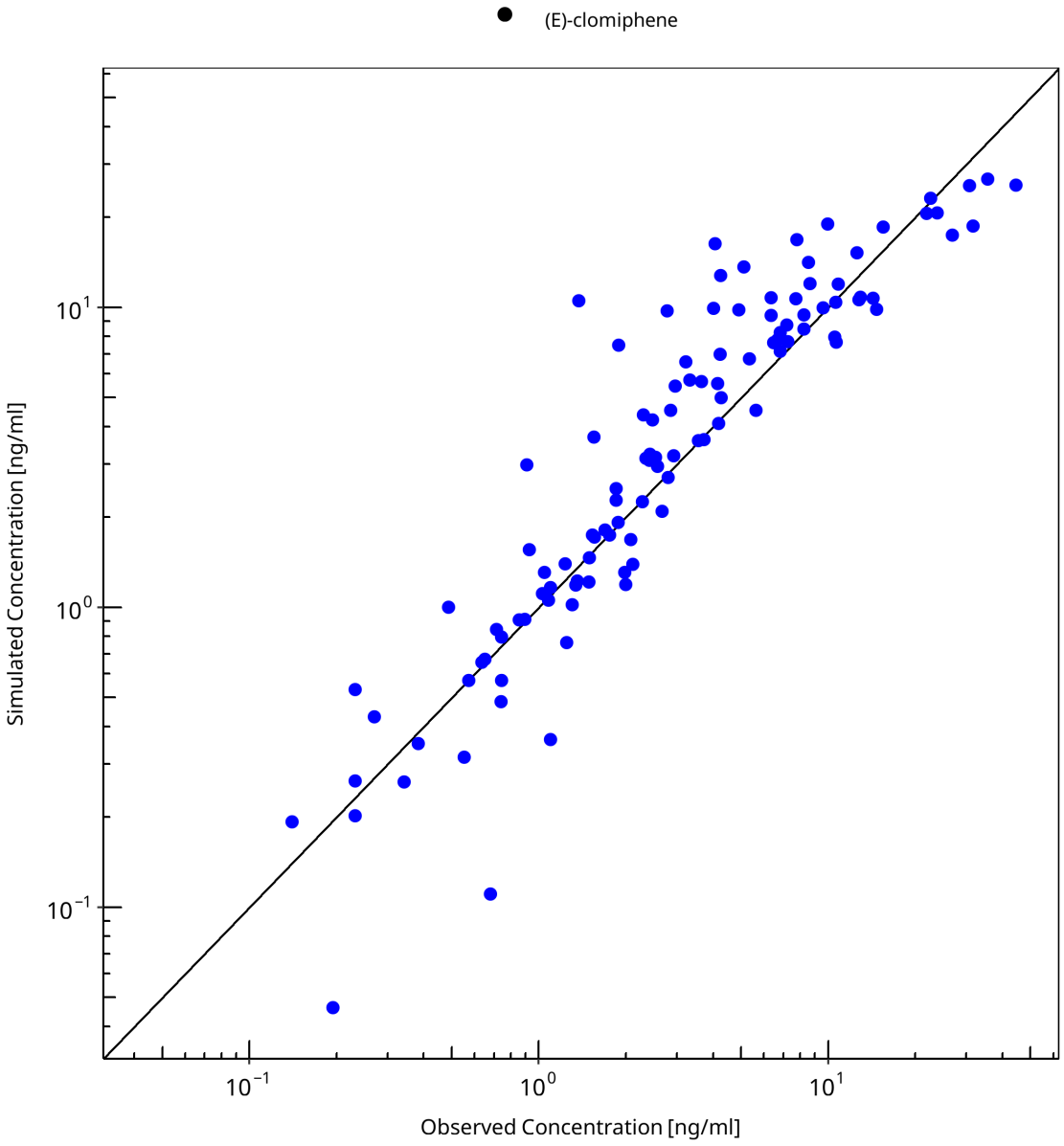


Figure 3-1: (E)-Clomiphene goodness-of-fit diagnostics

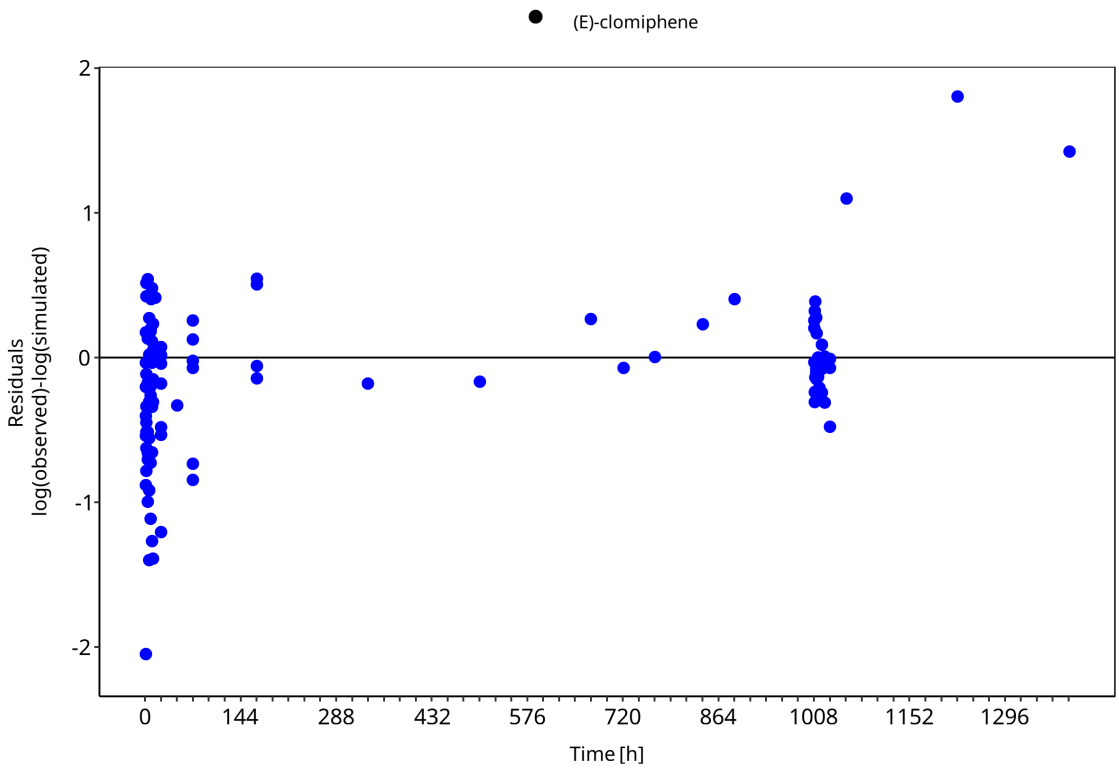


Figure 3-2: (E)-Clomiphene goodness-of-fit diagnostics

3.2.2 (E)-4-Hydroxyclomiphene goodness-of-fit diagnostics

(E)-4-Hydroxyclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This metabolite is directly informative for CYP2D6-dependent hydroxylation of (E)-clomiphene.

Table 3-2: GMFE for (E)-4-Hydroxyclomiphene goodness-of-fit diagnostics

Group	GMFE
(E)-4-hydroxyclomiphene	1.64

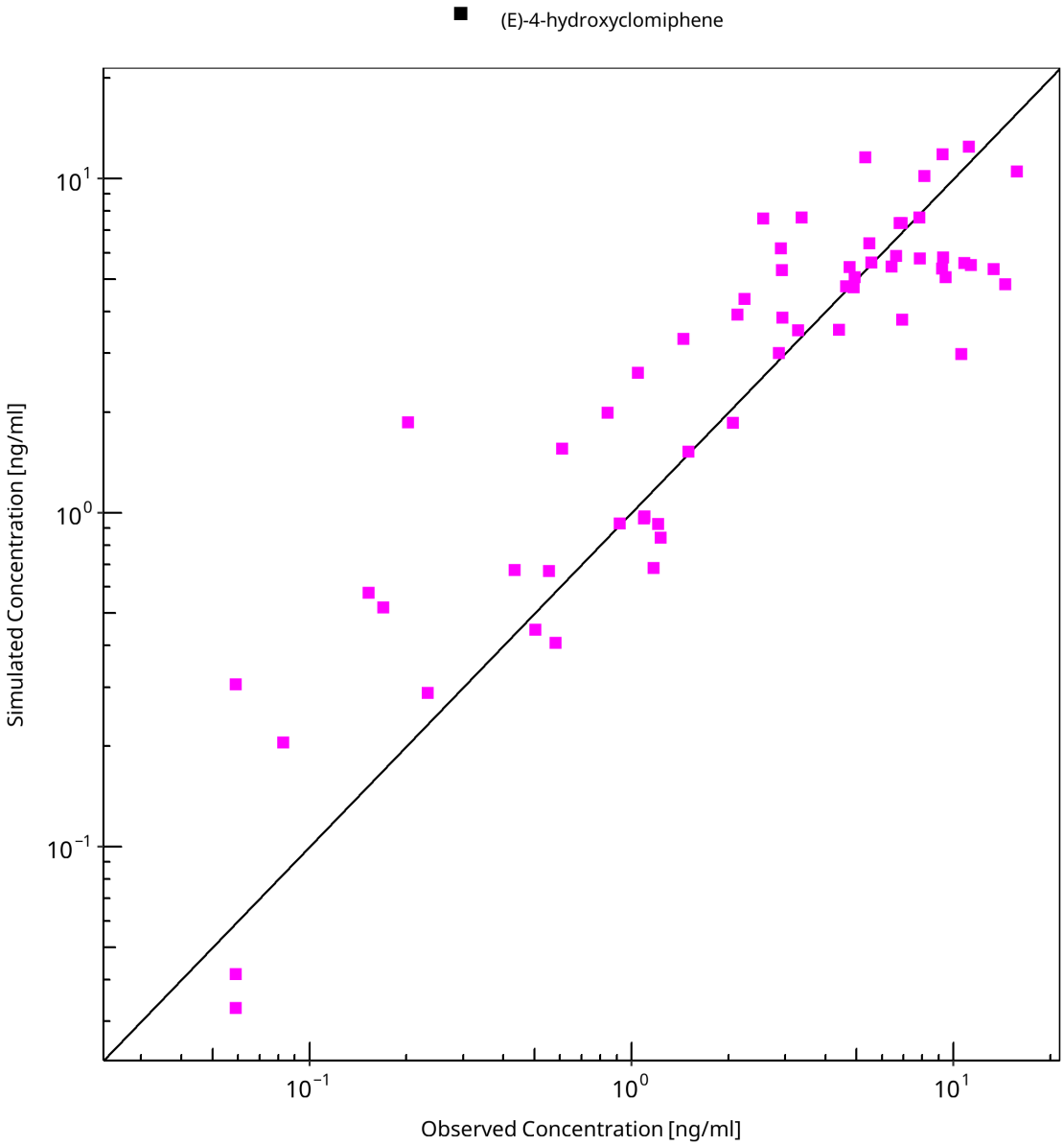


Figure 3-3: (E)-4-Hydroxyclomiphene goodness-of-fit diagnostics

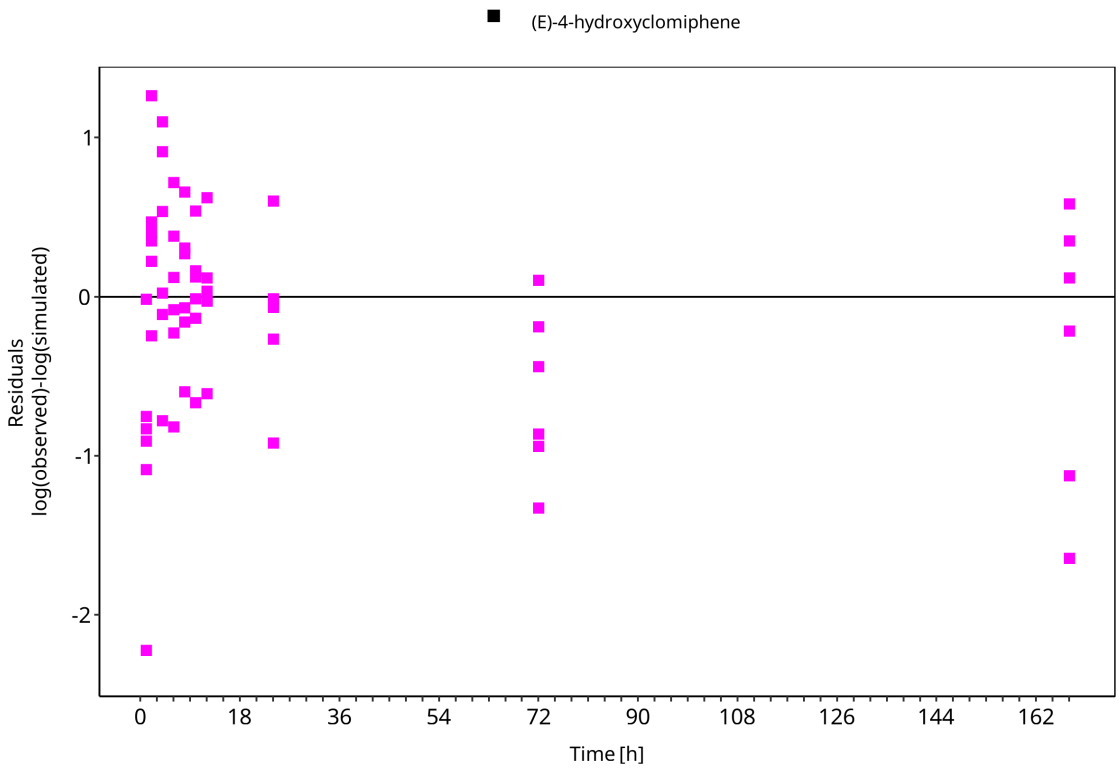


Figure 3-4: (E)-4-Hydroxyclomiphene goodness-of-fit diagnostics

3.2.3 (E)-N-desethylclomiphene goodness-of-fit diagnostics

(E)-N-Desethylclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This metabolite reflects desethylation contributions from CYP2D6 and CYP3A4 pathways.

Table 3-3: GMFE for (E)-N-desethylclomiphene goodness-of-fit diagnostics

Group	GMFE
(E)-N-desethylclomiphene	1.87

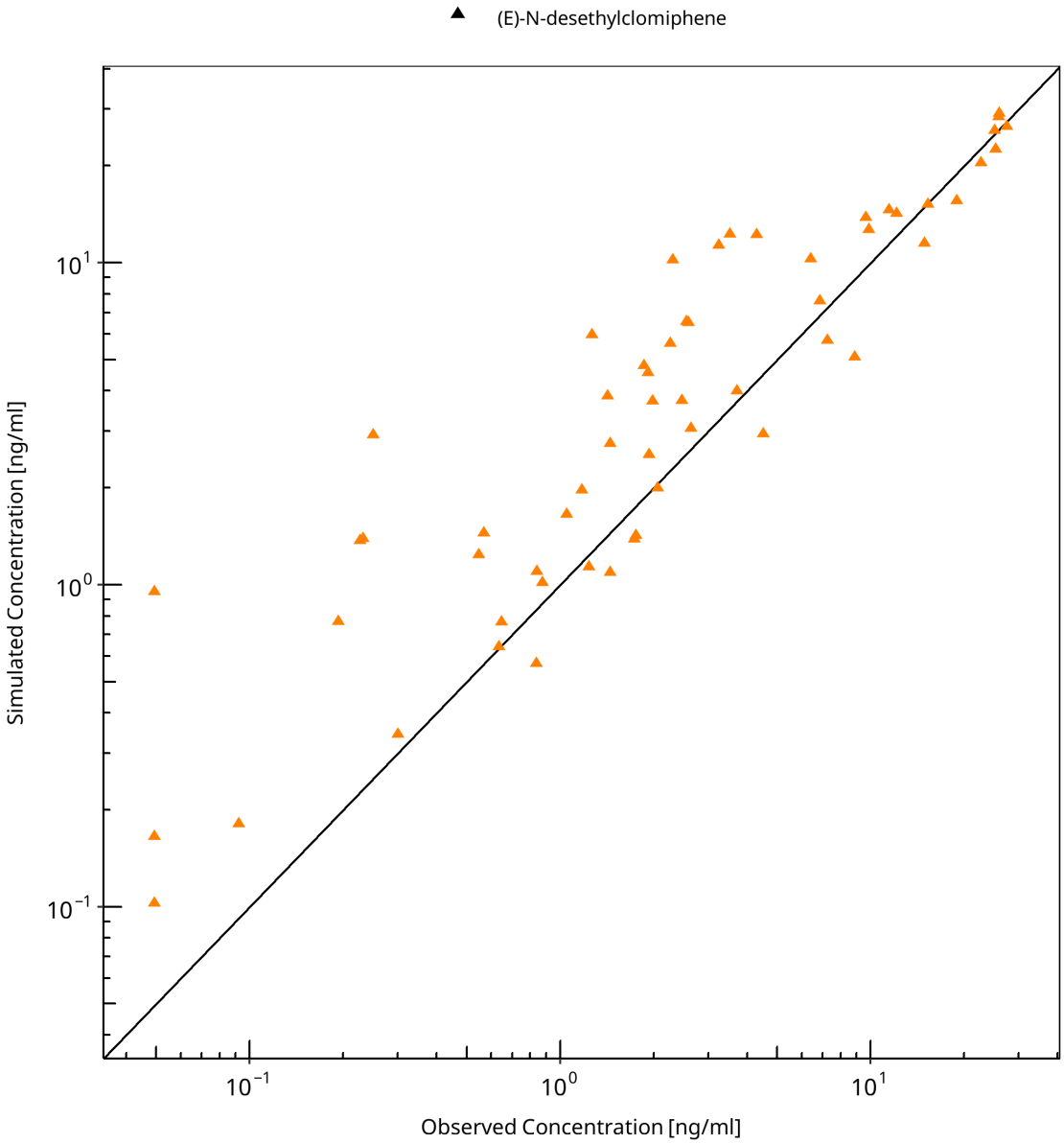


Figure 3-5: (E)-N-desethylclomiphene goodness-of-fit diagnostics

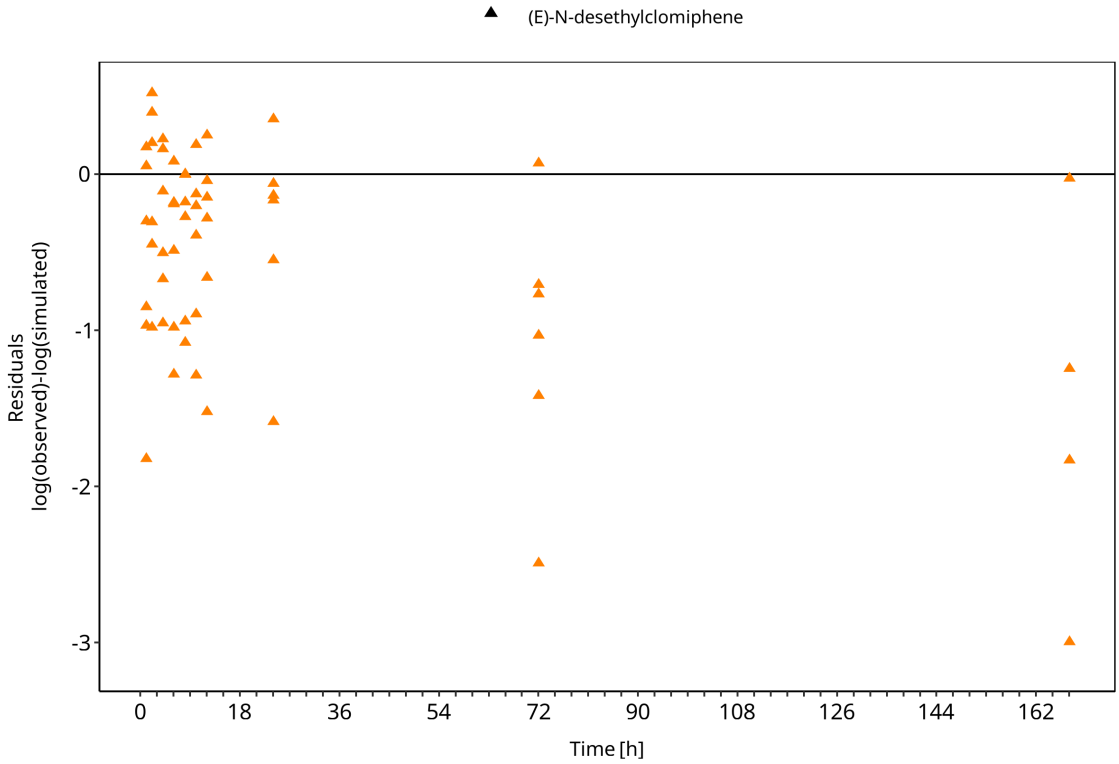


Figure 3-6: (E)-N-desethylclomiphene goodness-of-fit diagnostics

3.2.4 (E)-4-Hydroxy-N-desethylclomiphene goodness-of-fit diagnostics

(E)-4-Hydroxy-N-desethylclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This section supports evaluation of downstream metabolite formation and elimination.

Table 3-4: GMFE for (E)-4-Hydroxy-N-desethylclomiphene goodness-of-fit diagnostics

Group	GMFE
(E)-4-hydroxy-N-desethylclomiphene	1.56

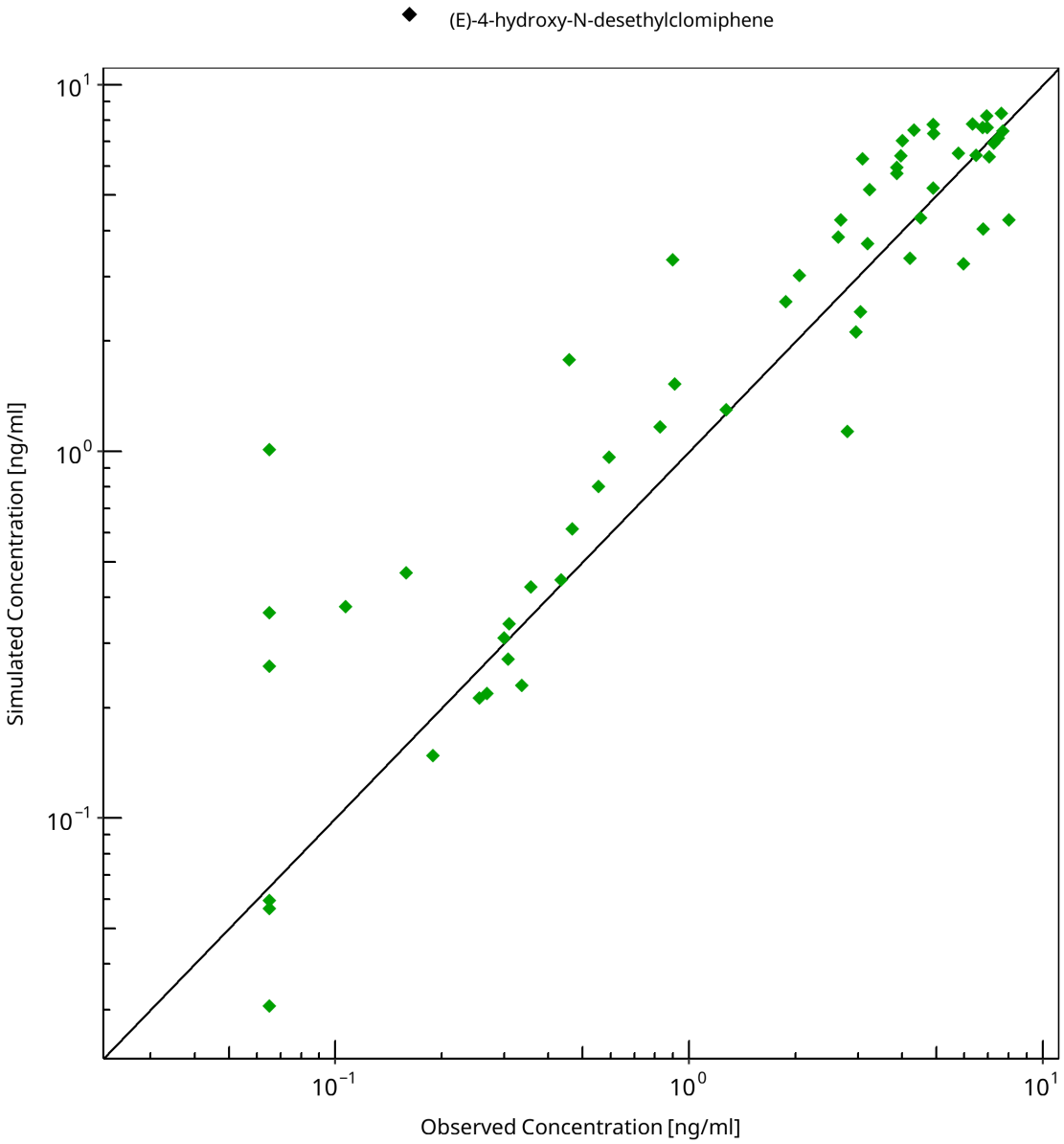


Figure 3-7: (E)-4-Hydroxy-N-desethylclomiphene goodness-of-fit diagnostics

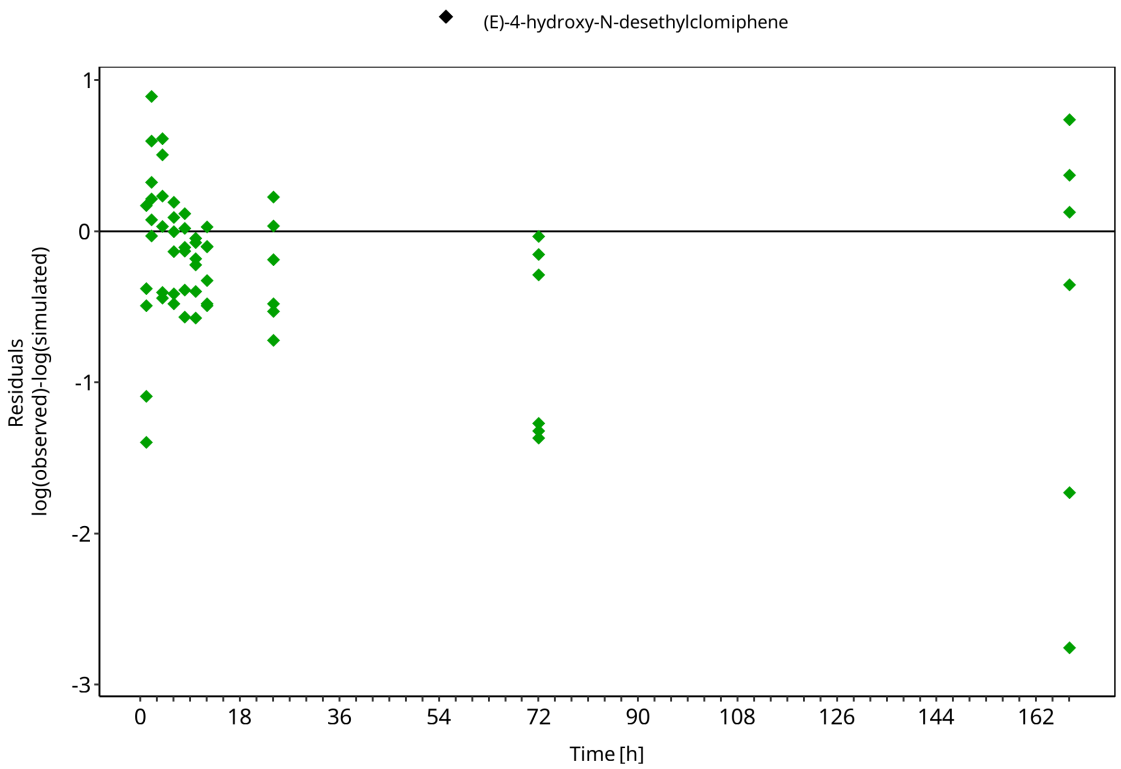


Figure 3-8: (E)-4-Hydroxy-N-desethylclomiphene goodness-of-fit diagnostics

3.3 Concentration-time profiles

The concentration-time profile sections show observed and simulated plasma concentrations for the CYP2D6 activity-score panel study and the external literature studies. Lines show simulated concentrations. Symbols show mean observed data. Error bars show the corresponding standard deviations, where available (Kovar 2022, Supplement Figures S1-S8). The section split follows the scientific role of the data rather than the analyte split used for GOF diagnostics.

3.3.1 Model Building

The model-building profiles comprise the AS = 0 and AS = 2 groups of the CYP2D6 activity-score panel with paired parent and metabolite observations.

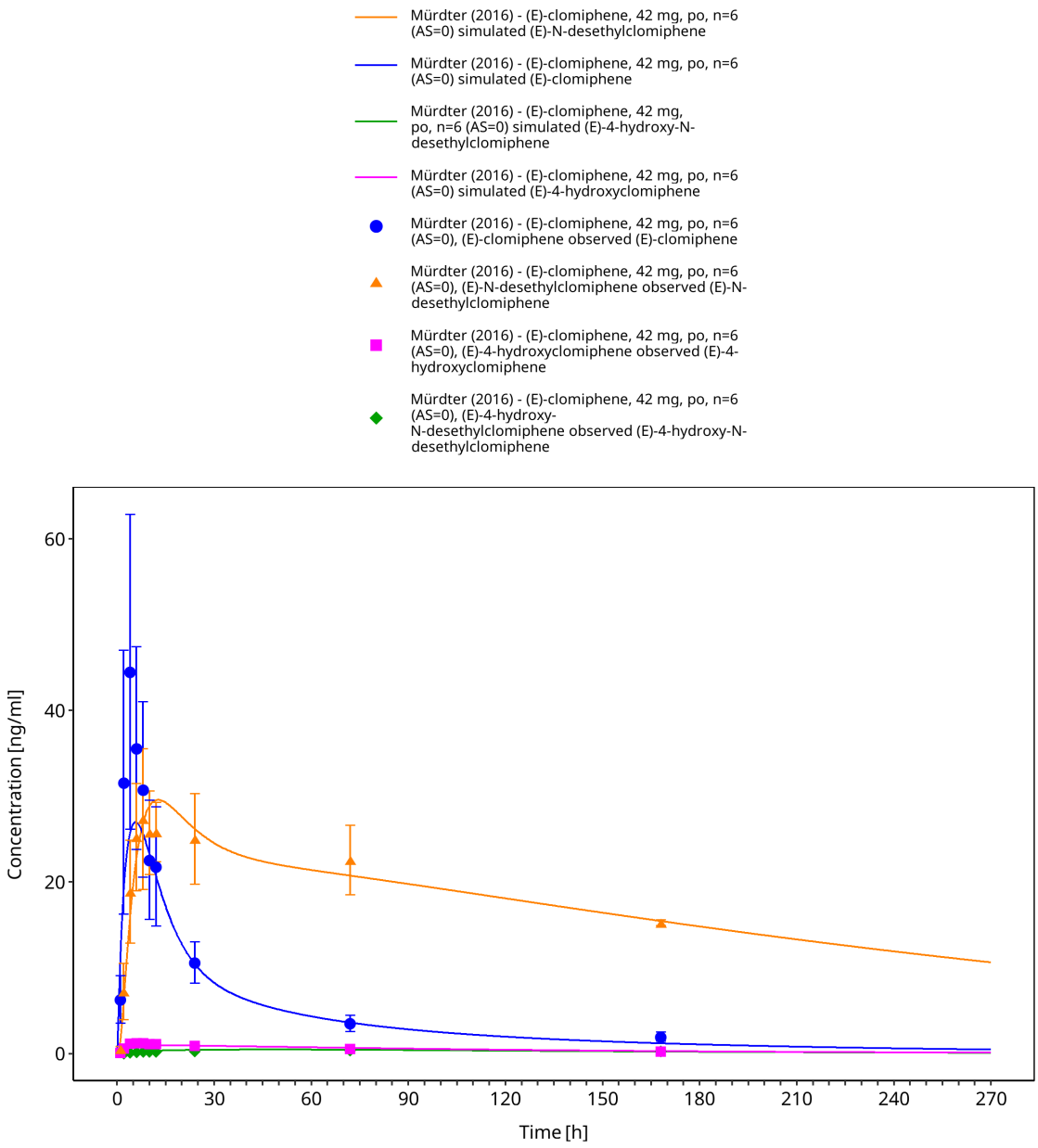


Figure 3-9: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=6 (AS=0)

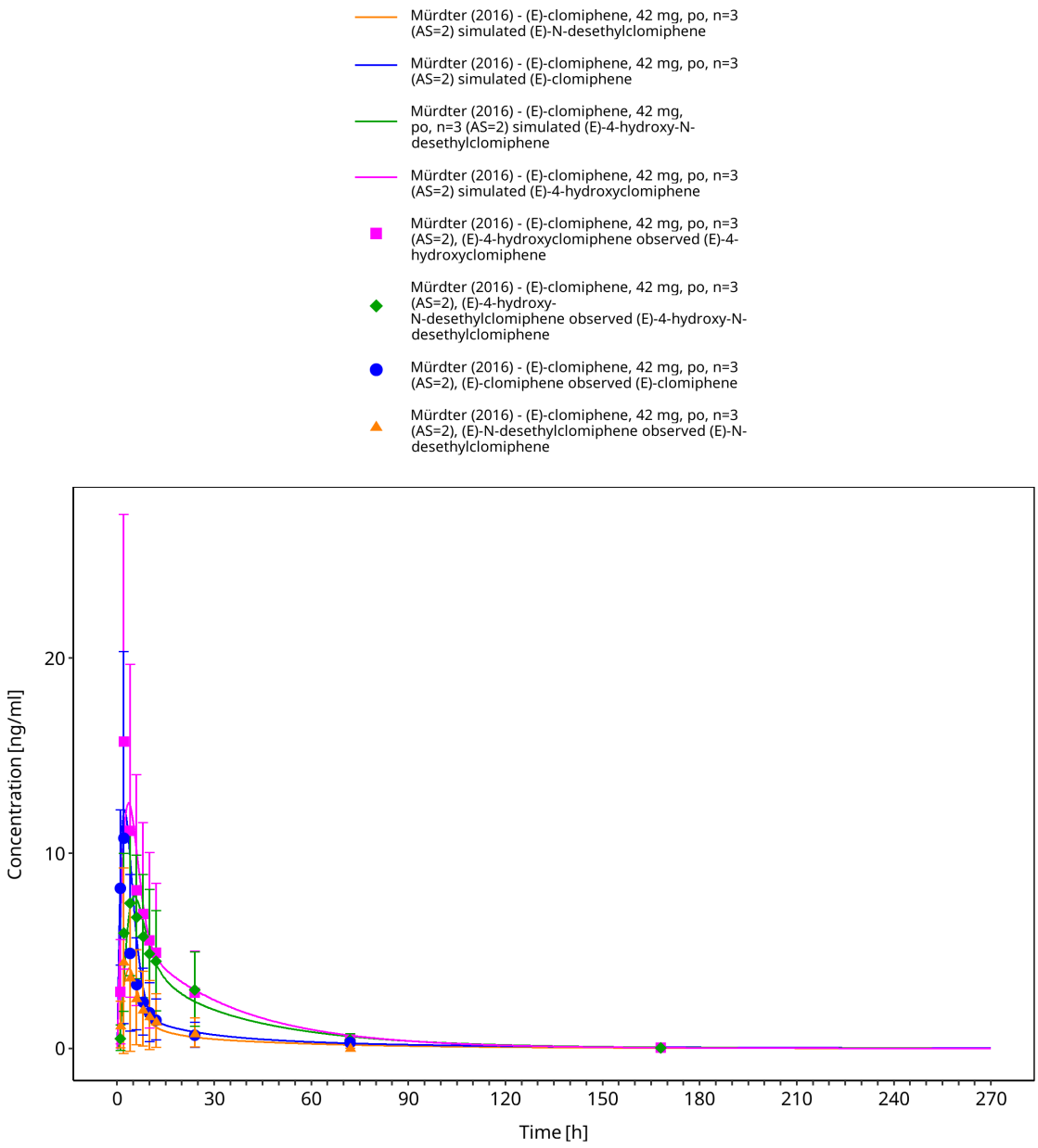


Figure 3-10: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=3 (AS=2)

3.3.2 Model Verification

The model-verification profiles comprise the AS = 0.5, 0.75, 1 and 3 groups of the CYP2D6 activity-score panel and literature studies after single-dose and multiple-dose oral administration. For each literature study, the CYP2D6 k_{cat} values were fitted to the observed profile (Table 1b).

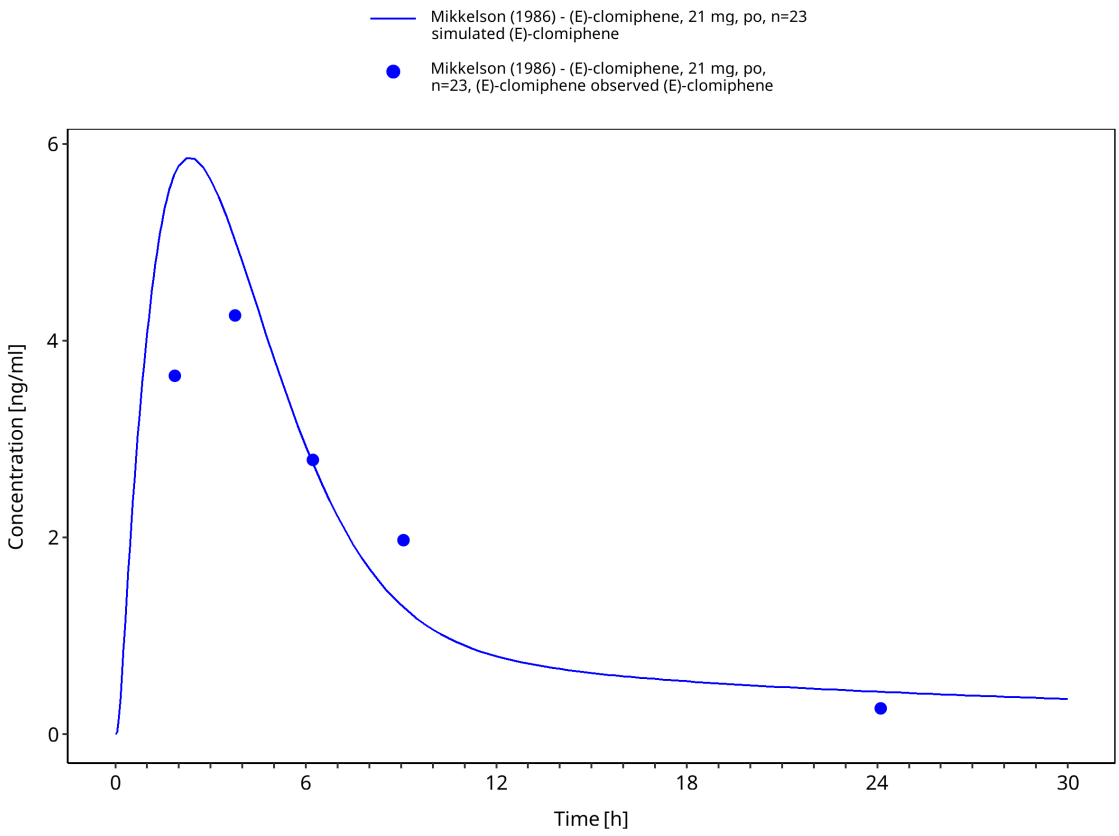


Figure 3-11: Mikkelson 1986: (E)-clomiphene, 21 mg, po, n=23

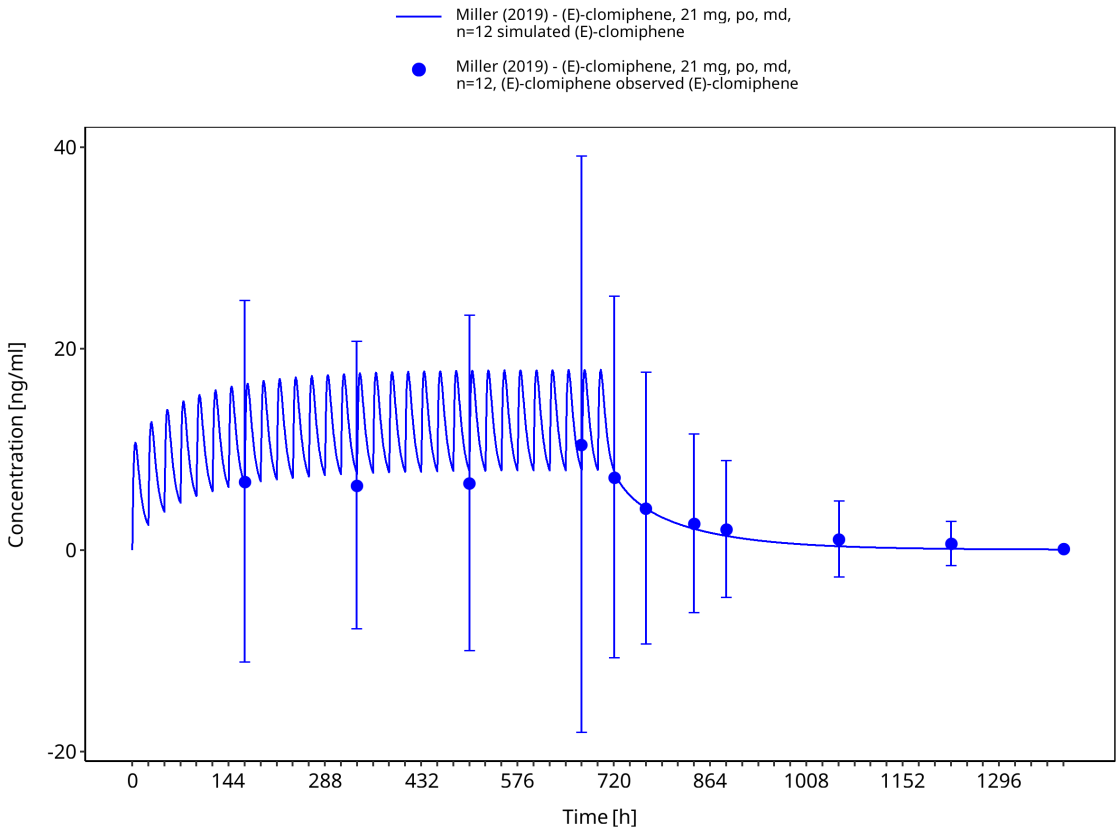


Figure 3-12: Miller 2019: (E)-clomiphene, 21 mg, po, md, n=12

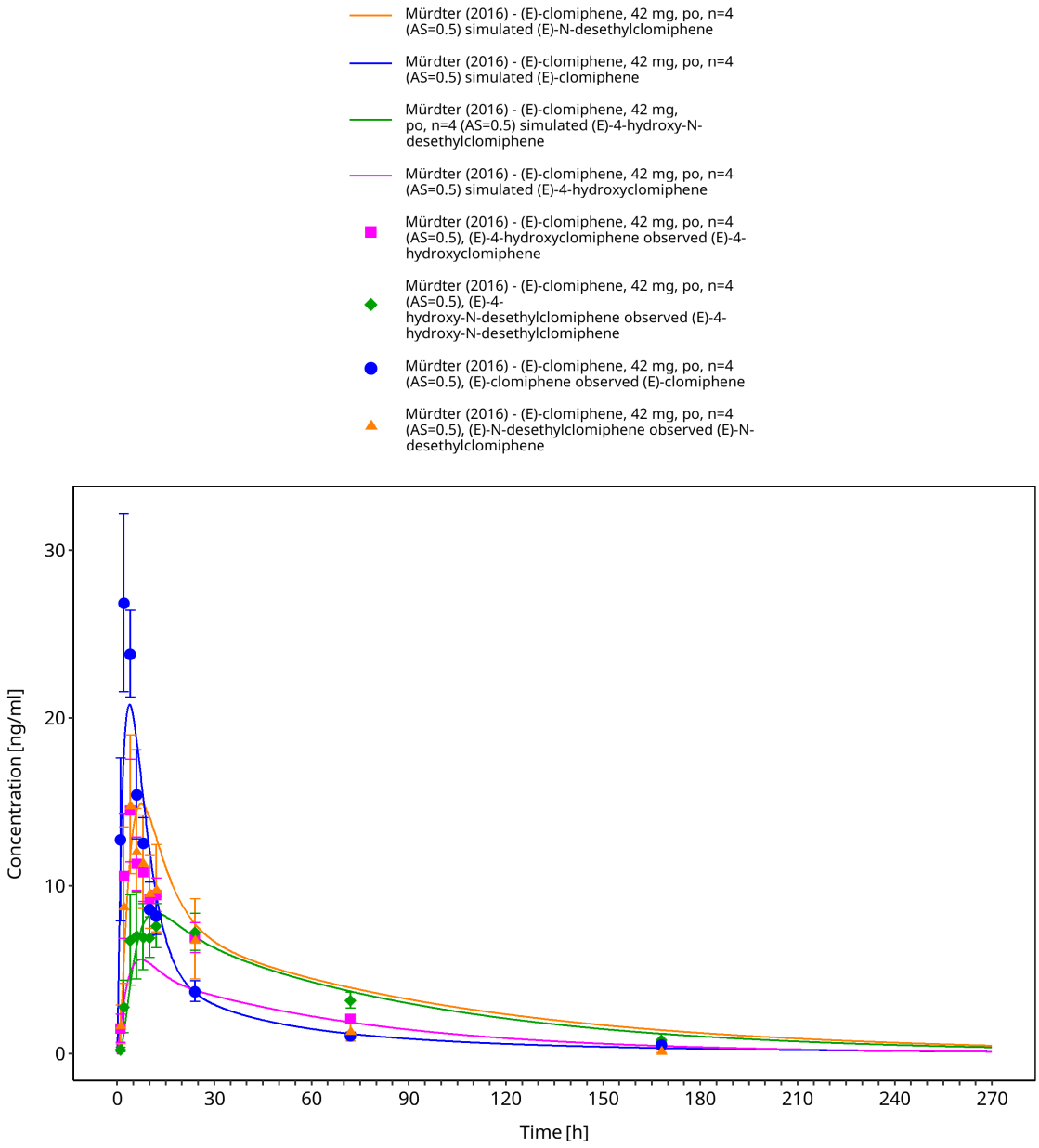


Figure 3-13: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=4 (AS=0.5)

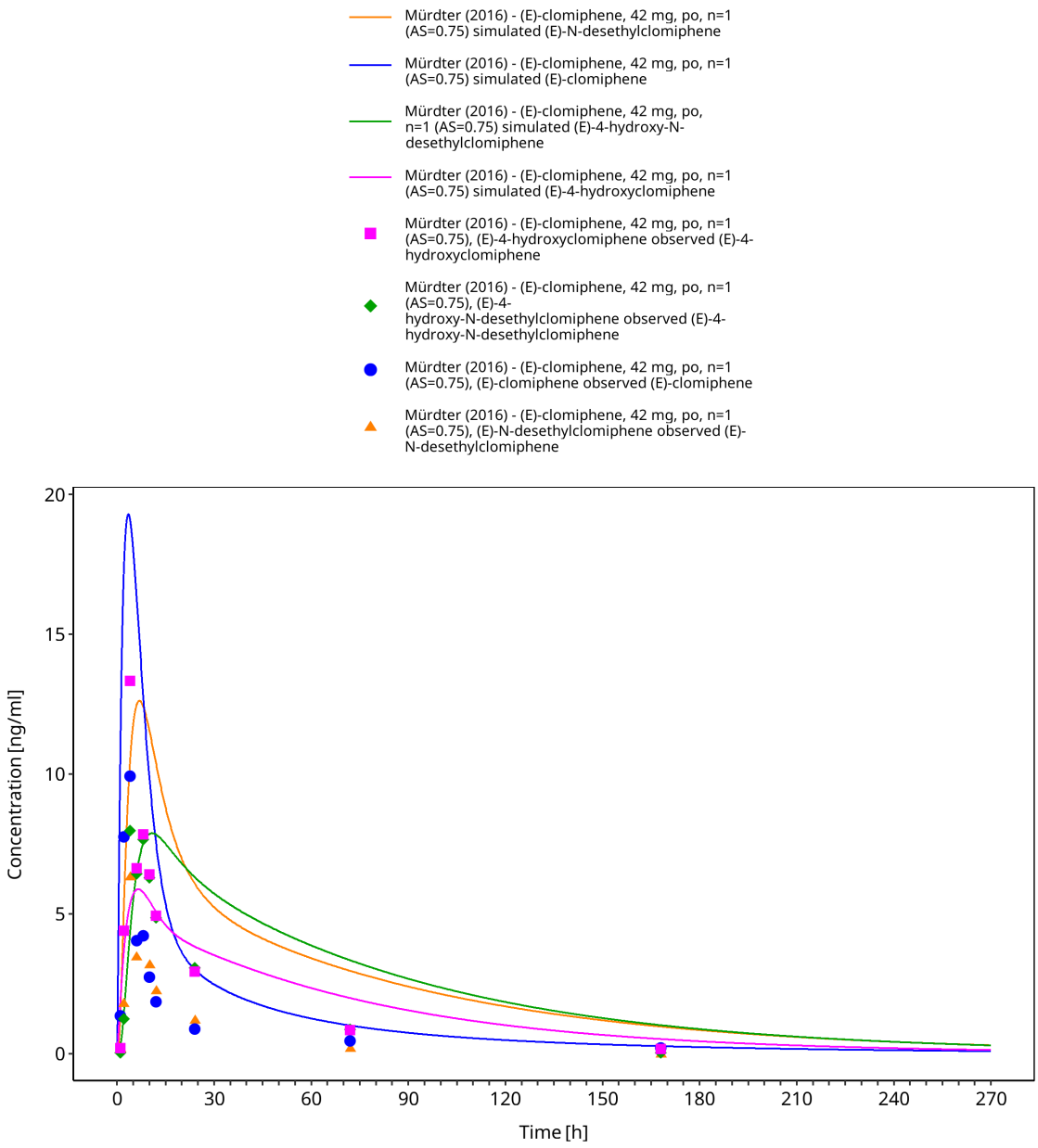


Figure 3-14: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=1 (AS=0.75)

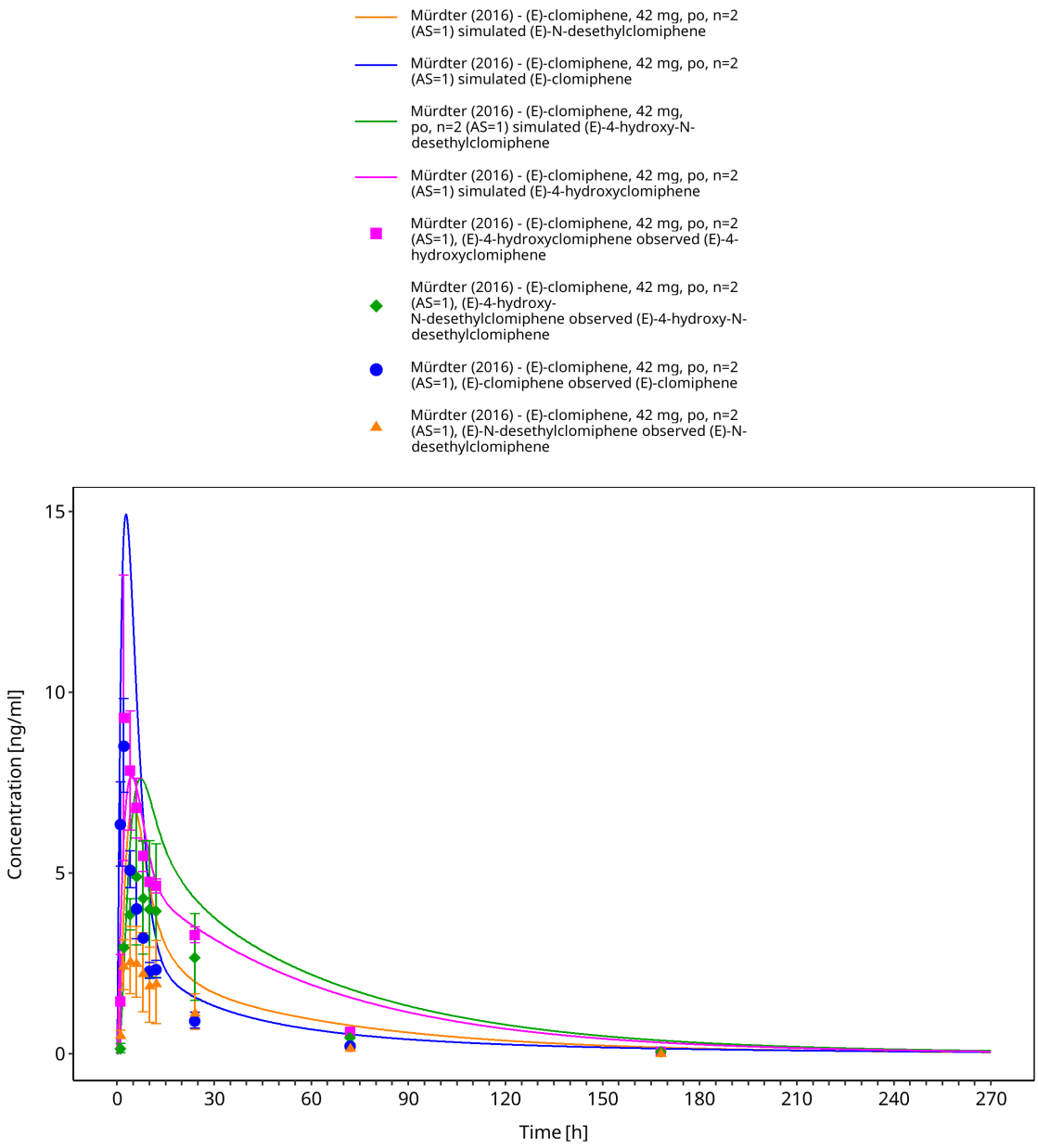


Figure 3-15: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=2 (AS=1)

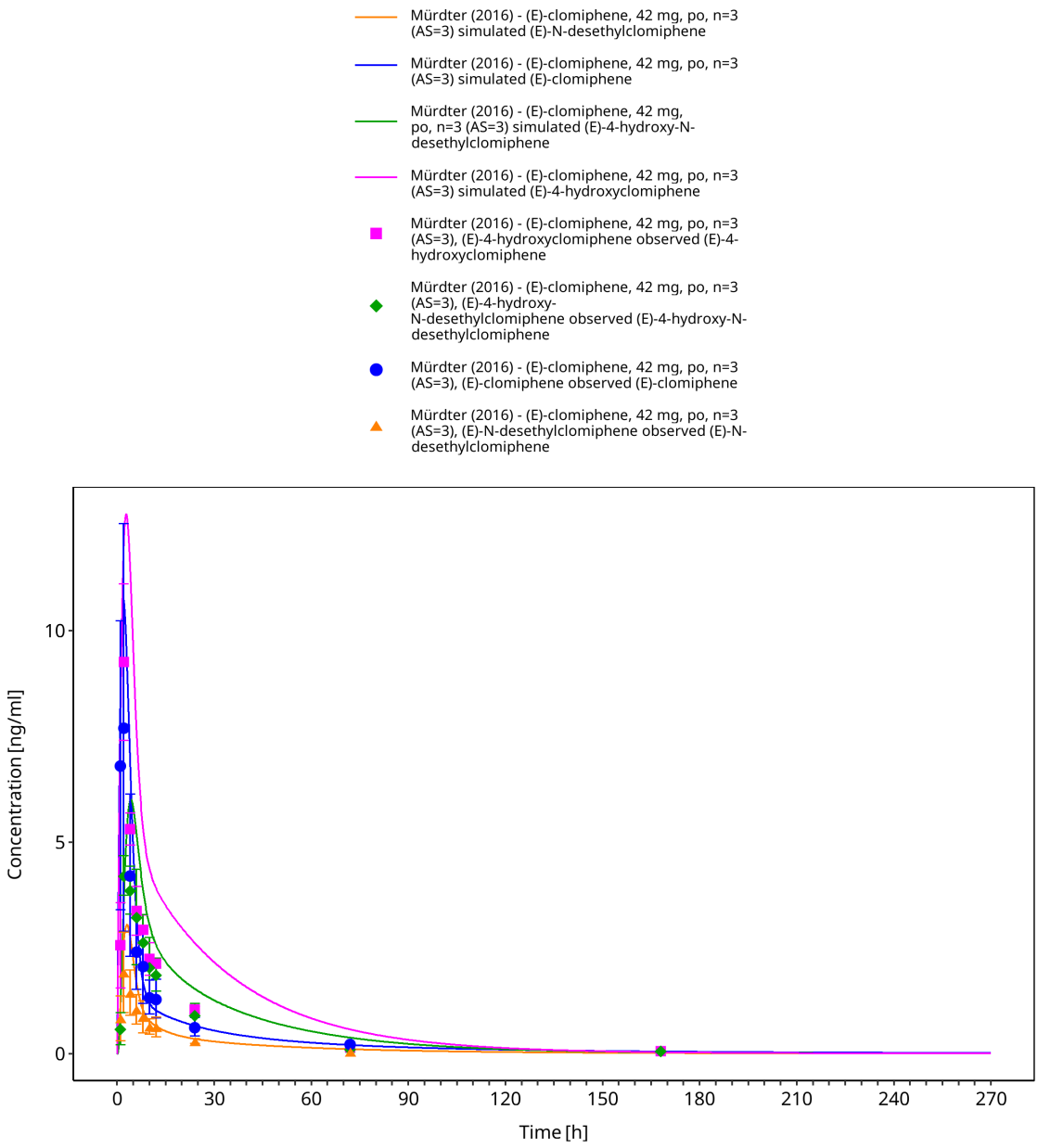


Figure 3-16: Mürdter 2016: (E)-clomiphene, 42 mg, po, n=3 (AS=3)

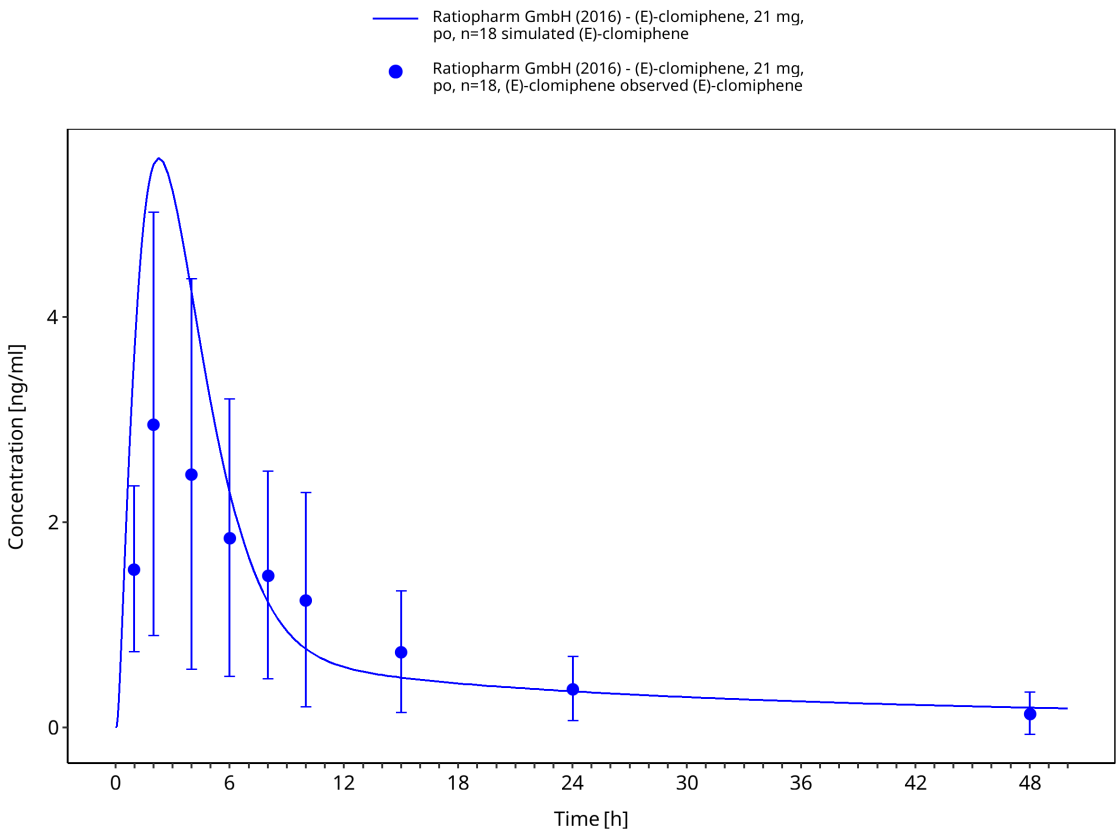


Figure 3-17: Ratiopharm GmbH 2016: (E)-clomiphene, 21 mg, po, n=18

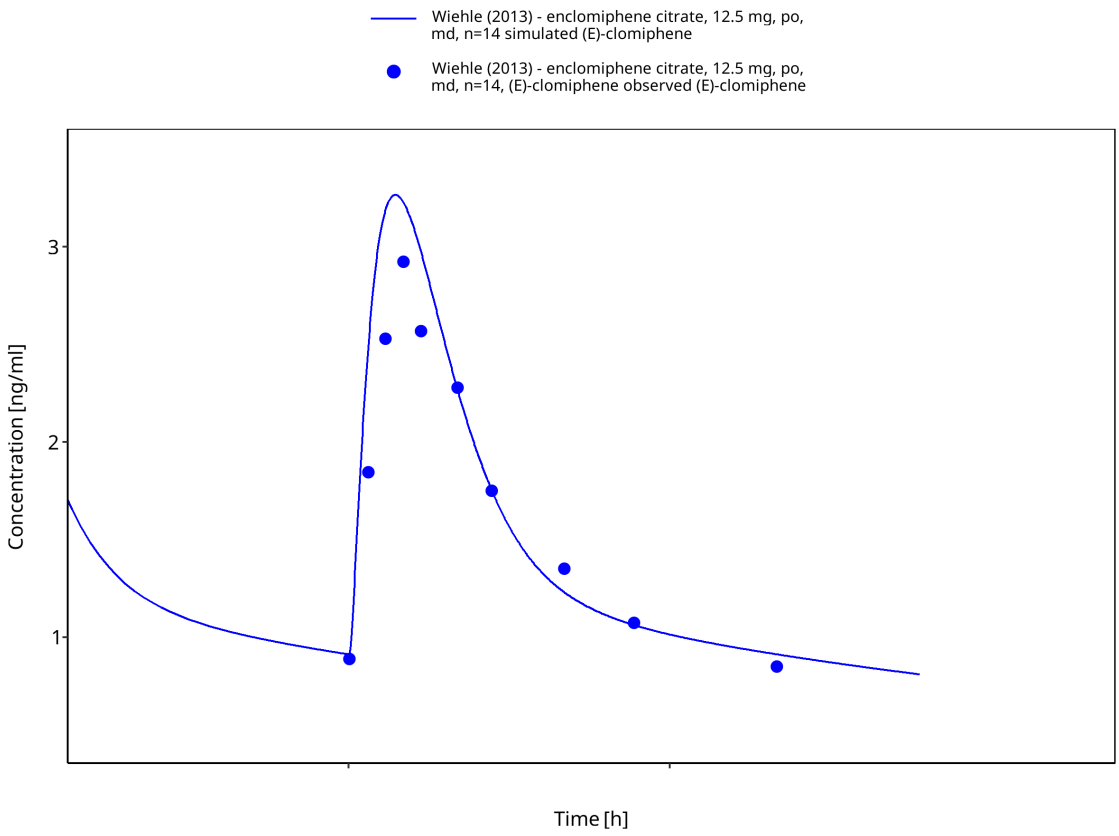


Figure 3-18: Wiehle 2013: enclomiphene citrate, 12.5 mg, po, md, n=14

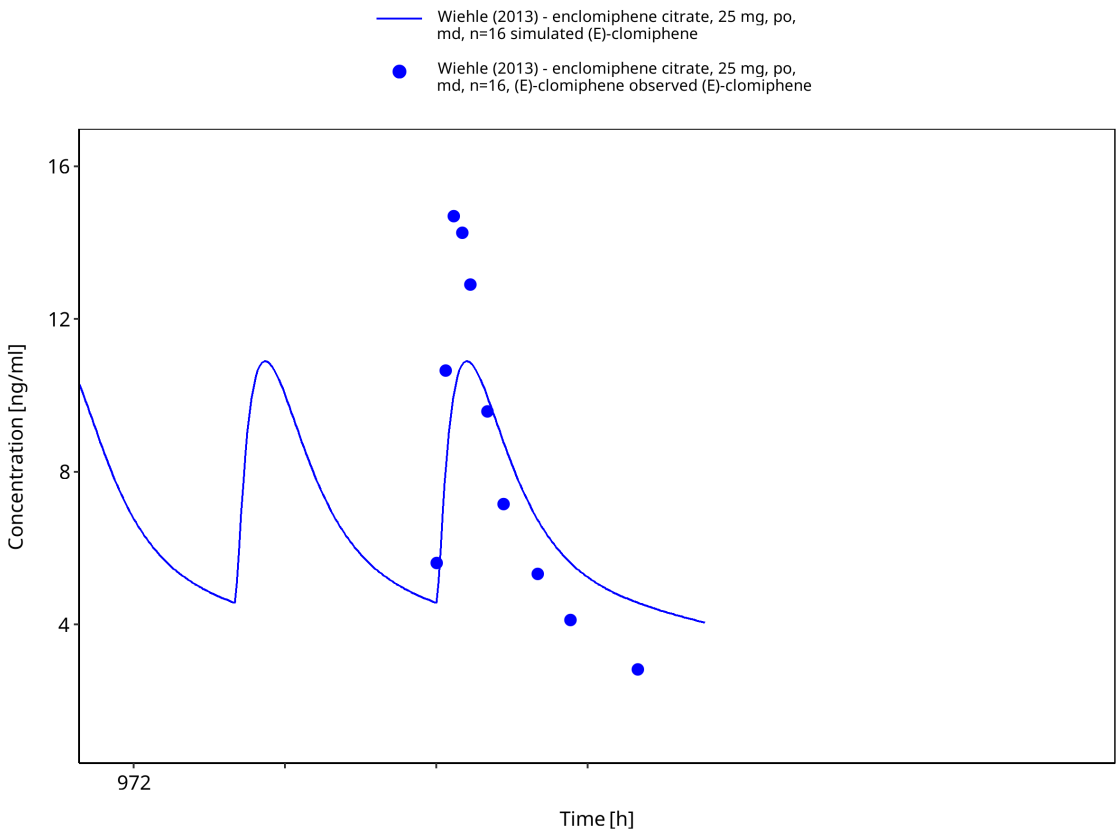


Figure 3-19: Wiehle 2013: enclomiphene citrate, 25 mg, po, md, n=16

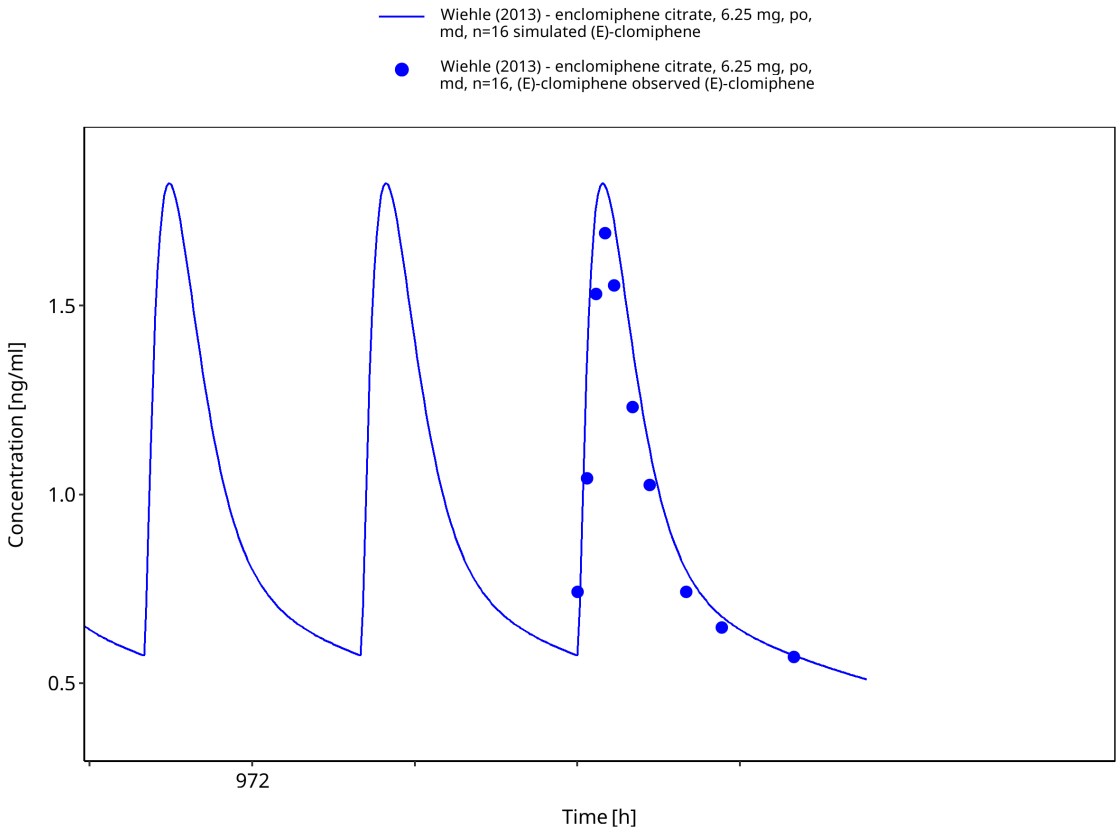


Figure 3-20: Wiehle 2013: enclomiphene citrate, 6.25 mg, po, md, n=16

4 Conclusion

The clomiphene PBPK model describes the evaluated plasma concentration-time data for (E)-clomiphene after oral administration in adults. Data for (E)-N-desethylclomiphene, (E)-4-hydroxyclomiphene, and (E)-4-hydroxy-N-desethylclomiphene come only from the single-dose Mürdter 2016 panel in healthy women. The evaluation covers the CYP2D6 activity scores 0, 0.5, 0.75, 1, 2, and 3. The single-dose and multiple-dose literature profiles were described after study-specific fitting of CYP2D6 k_{cat}.

The model uses common absorption and distribution assumptions across CYP2D6 groups. Activity-score-dependent CYP2D6 catalytic rates govern hydroxylation, desethylation, and downstream metabolite elimination. CYP3A4, CYP2B6, passive renal filtration, residual hepatic clearance, and enterohepatic recirculation provide additional disposition pathways.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated dose, regimen, analyte, population, and CYP2D6 ranges. No formal acceptance criterion was applied.

The main limitations are the small intermediate-metabolizer groups, extrapolated CYP2D6 catalytic rates, and limited paired parent-metabolite data outside the activity-score panel. The literature profiles used CYP2D6 k_{cat} values that were fitted to each profile. They are therefore not independent evidence for multiple-dose predictions or for populations without CYP2D6 information. The study-specific k_{cat} values can also compensate for other differences between studies, for example in formulation, regimen, or analyte identity.

5 References

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